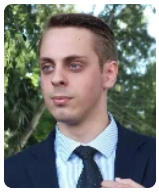
Petar Lukovic

COMPUTER VISION · MSC ELECTRICAL ENGINEERING, ETH ZÜRICH

Zürich, Switzerland petar.lukovic032@gmail.com LinkedIn Website



PROFESSIONAL SUMMARY

Electrical Engineering MSc (ETH Zürich) focused on Computer Vision, 3D Vision and Robotics, with research experience in SLAM, visual localization and mixed reality. Communicative team player, seeking opportunities in these fields.

EDUCATION

Sept 2023 — Present **Master's, ETH Zürich** Zürich
- Department of Information Technology and Electrical Engineering (D-ITET), specialization in Signal Processing and Machine Learning

Oct 2019 — Jul 2023 **Bachelor's, University of Belgrade** Belgrade
- School of Electrical Engineering, Department for Signals and Systems

EXPERIENCE

Mar 2025 — Present **Research Assistant, ETH Zürich** Zürich
- Computer Vision and Geometry Lab (CVG)

Sept 2025 — Feb 2026 **Teaching Assistant, ETH Zürich** Zürich
- Teaching Assistant on Mixed Reality (HS25) at Computer Vision and Geometry Lab

Sept 2024 — Feb 2025 **Teaching Assistant, ETH Zürich** Zürich
- Teaching Assistant on Computer Vision (HS24) at Computer Vision and Geometry Lab

Oct 2022 — Jun 2023 **Teaching Assistant, School of Electrical Engineering** Belgrade
- Teaching Assistant on Object-oriented Programming (HS20) and Microcontroller Systems (HS22)

PROJECTS AND PUBLICATIONS

Mar 2026 — Sept 2026 **Polygraph: Large-Scale Multi-Session Trajectory Optimization Using Pose Graphs. Master Thesis, Prof. Dr. Marc Pollefeys** CVG, ETH Zürich
- Mapless multi-session trajectory optimization with pose graphs and learned metric depth. Under submission @ ICRA 2027

Mar 2025 — Present **CroCoDL: Cross-device Collaborative Dataset for Localization** CVG, ETH Zürich
- Data processing and maintenance. Challenge organiser @ ICCV 2025. Website Challenge

Oct 2024 — Mar 2025 **Spot-On: A Mixed Reality Interface for Multi-Robot Cooperation** CVG, ETH Zürich
- Mixed reality interface for human–robot collaboration. CV4MR Workshop @ CVPR 2025. Paper

Sept 2024 — Mar 2025 **HoloSpot: Intuitive Object Manipulation via Mixed Reality Drag-and-Drop** CVG, ETH Zürich
- Integration of Boston Dynamics Spot robot and Microsoft HoloLens 2 for Drag and Drop tasks with help of OpenMask3D and AnyGrasp. Website

Mar 2024 — Aug 2024 **Multiple to One Matching. Semester project, Prof. Dr. Marc Pollefeys** CVG, ETH Zürich
- Matching multiple database images to one query for camera localization and 3D reconstruction (HLoc)

Aug 2025 — Dec 2025 **Interactive 3D Segmentation. Semester project, Prof. Dr. Siyu Tang** VLG, ETH Zürich
- Interactive 3D Segmentation using Segment Anything

Mar 2023 — Jul 2023 **Stereo Vision Depth Estimation. Bachelor Thesis, Prof. Dr. Dragomir El Mezeni** Belgrade
- Theoretical basis for Stereo Depth estimation. Practical implementation on Nvidia Jetson Nano. Thesis

EXTRACURRICULAR ACTIVITIES

Oct 2021 — May 2023 **Team member, Formula Student team Road Arrow** Belgrade
- Member of the team for vehicle dynamics and powertrain

LANGUAGES

English C2 Serbian Native

SKILLS

Python, C / C++, CUDA
Unity, Linux, Git
SLAM, Robotics, Computer Vision, 3D Vision
HoloLens 2, Meta Quest 3, Aria Gen 2
Boston Dynamics Spot